

Description

Hybrid in-office in Auckland, NZ

Timescapes is looking for a Machine Learning Engineer to join our small, but growing team of high-performers. We're after someone who approaches problems from first principles, honing in on underlying causes and diligently iterating towards the best solution. We believe that great products feature incredible experiences that customers never want to stop using.

About us

Founded in New Zealand in 2017, Timescapes is a visual progress tracking solution for complex construction projects – our mission is to simplify construction through shared visibility. Customers value Timescapes because it helps them stay on schedule, validate construction claims and communicate progress more effectively.

We're a rapidly growing company, and are used by some of the largest construction firms across Australia, Canada, New Zealand and the United States. We're deliberate and thoughtful in everything that we do, and we emphasise awareness, speed and quality in our approach to product development.

About the role

As our second dedicated ML / AI hire at Timescapes, you'll be working closely with our product and engineering team to develop ML models (computer vision) and incorporate them into our existing (and new) product capabilities. Key aspects of the role include:

- Working closely with other members of the AI / ML team to design, develop, integrate and support machine learning models, to solve specific construction problems related to tracking activity and progress on civil, institutional and commercial sites.
- Running experiments with LLMs and other large-scale generative AI models during problem and solution exploration, and determining when to invest in custom development vs leveraging off-the-shelf models.
- Developing software to manage the ML lifecycle, including data management, labeling and training.
- Sourcing training data, curating it for quality, and determining how best to integrate it into the ML lifecycle.

Essential Skills / Qualifications

- Bachelor's degree in Computer Science, Engineering, or a related field.
- Experience with the development and delivery of production-level software systems that feature critical machine learning components.
- Experience with the development of computer vision models, on both the data and model sides.
- Experience with end-to-end ML operations, including data acquisition, labeling, model training, pipeline design + management, continuous integration, model versioning, and performance monitoring.

- Experience with leveraging LLMs, LVMs and multimodality models to augment product capabilities, including strong experimentation skills, and the ability to combine off-the-shelf commercial model capabilities with custom ML models.
- Familiarity with methods for ML model validation and verification.
- Experience deploying ML inference infrastructures for systems with high throughput.
- Ability to develop ML-enabled product capabilities in the context of the entire SDLC, from initial concepts through detailed design, development, deployment and maintenance.

Other Preferred Skills / Qualifications

- Experience with deep learning frameworks e.g. PyTorch
- Familiarity with inference engines e.g. TensorRT
- Experience with developing edge ML applications
- Experience writing with .NET, C# and Python

About you

We want curious, driven and enthusiastic people who are motivated by the prospect of building solutions that endure. The people who thrive at Timescapes are critical thinkers who are not afraid to embrace the unconventional. If the following resonates with you, we should definitely chat:

- **Value-oriented:** we're focussed on ensuring that Timescapes as a business gets genuine value from engineering investments. You'll be someone who sees methodologies, best practices and patterns as useful tools, but you don't blindly adopt them and never let them get in the way of great customer outcomes.
- **Obsessed with the "why":** you're someone who cares about the details, and you take pride in sharing your rationale for every decision you make. You question things that don't make sense, and you expect others to do the same.
- **All about the editing:** when it comes to great solutions, we don't believe you generally nail it on the first attempt. You're someone who's not afraid to leave a lot of ideas on the cutting room floor, and you intuitively understand that the first release is just the beginning.
- **Adaptable and creative:** as a small company doing ambitious things, we don't always have the right answer, and we know that what's worked in the past may not work in the future, but we're not deterred by the unknown, and we're excited about trying new things and finding our way through.
- **Communication matters** - in addition to communicating clearly and succinctly, you're also a natural relationship-builder, and you're equally comfortable on Slack/email, video calls and in person.

Benefits

- Flexible work hours
- Four weeks leave each year
- Health insurance
- 5% employer Kiwisaver contribution

Sound like you?

If the prospect of working with talented people on challenging problems in a globally-expanding tech company appeals to you, let us know below.

Recruitment Process

Round	Description	Purpose	Participants
1	Recruitment screen	Qualify candidate, gauge comfort with range of responsibilities.	Recruiter
2	First interview	Provide context, understand career history, answer questions	Shehan
3	Technical interview	Assess problem solving ability and overall suitability for the role.	Qinglin, Trevor
4	Technical exercise	Assesses ML-specific engineering and solution design skills.	Qinglin, Tom
6	Culture fit	Behavioural interview to determine alignment with Timescapes values.	Tim, Jono